Who Bears the Burden of Bank Capital Regulation?

Buffers, Risk Weights, and the Supply of Credit

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Motivation

Why capital requirements exist: equity absorbs unexpected losses, so banks stay solvent in downturns.

The debate over how far they should increase has been conducted in aggregates:

stability benefits of a higher requirement vs. the output cost of getting there

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The debate over how far they should increase has been conducted in aggregates:

stability benefits of a higher requirement vs. the output cost of getting there

But the regime is not an aggregate lever—it is deliberately differentiated:

- Risk weights: Business (C&I) loans $\omega_B = 1.00$ vs. mortgages $\omega_M = 0.50$
- Size tiers: asset thresholds at \$10B and \$50B over the sample — revised in 2019

Research Question

Who bears the burden of a capital requirement tightening?

- Which **sectors** absorb the contraction in credit?
- Which **banks** transmit it?
- What is the **aggregate cost**—and does design matter for it?

What I Do

Empirics: measure who bears a tightening

- US Call Reports, 4,800 banks, 1994–2019; identified capital-requirement shocks; within-bank local projections
- Two margins: **across *sectors*** (C&I vs. real estate), and **across *banks*** by distance to their own requirement

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Quantitative model: the same two margins, solved

- A continuum of banks, each choosing its own distance to its requirement; the capital constraint binds occasionally
- Estimated by SMM on the *sectoral* responses only — the cross-section is left free

Technique: policy rules and shadow prices parameterised by deep neural networks

What I Find

Empirics: two facts

1. Across sectors: business (C&I) lending contracts **1.3**× as much as real estate

2. Across banks: banks with the least room above *their own* requirement cut **1.3**× as much as those with the most

Quantitative model: fitted to fact 1, it reproduces fact 2

- **Across banks: 1.2**× in the model vs. **1.3**× in the data — a moment the estimation never used

The burden falls on banks with the least room above their own requirement.

Level of requirements ⇒ aggregate cost. *Design* ⇒ **incidence**.

What's New: the Buffer Distribution

What: each bank's ***distance to its own requirement*** — the buffer $D_{i,t} \equiv \mathrm{CR}_{i,t} - \kappa_{i,t}$, in percentage points

- What it costs a bank to raise wholesale funding (γ_i) makes that distance a *choice*: **costlier funding, thicker *target* buffer**

- **This paper solves at every point of the buffer distribution — not one intermediary, not two types**

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Context: the closest quantitative models do not carry that distribution (Elenev, Landvoigt, and Van Nieuwerburgh, 2021; Corbae and Erasmo, 2021)

Proposition 1 Related lit.

Roadmap: the Paper and This Talk

The paper has seven sections. This talk covers five.

1. Introduction ✓
2. Empirical Evidence ✓
3. A Stylized Model
4. Quantitative Model ✓
5. Quantitative Results ✓
6. Discussion
7. Conclusion ✓

Funding Capacity and the Buffer

- **Buffer** $D_{i,t} \equiv \mathrm{CR}_{i,t} - \kappa_{i,t}$ (distance to its own requirement, pp)

  $\mathrm{CR}_{i,t}$: capital ratio (regulatory capital / RWA). $\kappa_{i,t}$: bank i's requirement (κ_t plus its statutory tier).

- **Funding Capacity** $\equiv$ (core retail + NIB deposits) / total liabilities

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Across banks of *similar size*, those that rely on wholesale funding hold *thicker* buffers; those with stable deposit franchises hold *thinner* ones.

Regulation | Distribution | Regression | Leverage vs. RWA | Buffer by size | Proposition 1

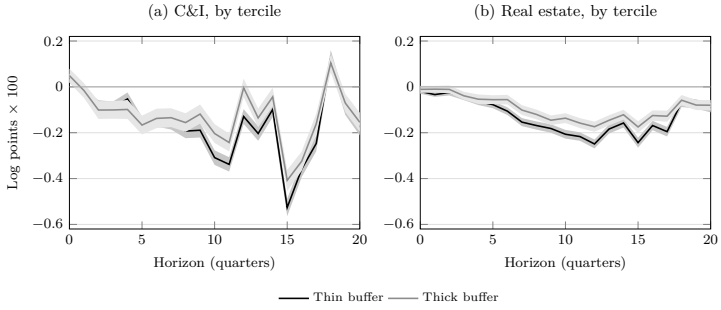
Two Empirical Facts

The Two Facts

Identification: narratively identified capital-requirement shocks (Drechsel and Miura, 2025); within-bank local projections

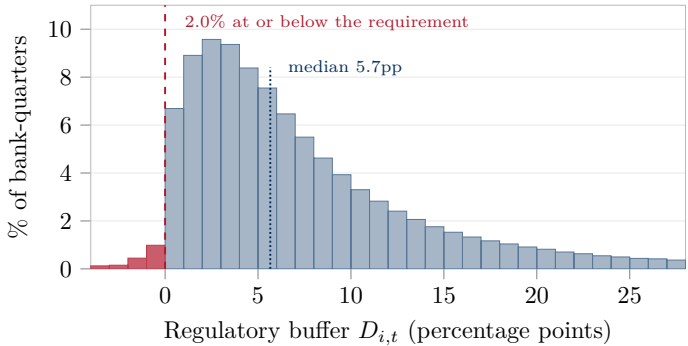
Fact 1. Across sectors: C&I contracts **1.31**× the real-estate response

Fact 2. Across banks: thin-buffer banks contract 1.32× more in C&I, 1.41× in real estate.



All four | Identification | Interpretation | Details | Precision

The Buffer Distribution in the Data



$D_{i,t} \equiv \mathrm{CR}_{i,t} - \kappa_{i,t}$. Call Reports, 1993Q2–2019Q4.

What the Buffer Gradient Identifies

A capital constraint alone cannot produce it.

- *Every* tercile contracts — even *thick*-buffer banks, furthest from their requirement, cut significantly (-0.13%)
- Yet fewer than 3% of bank-quarters are ever *at* their requirement — so a binding constraint cannot be what transmits the tightening

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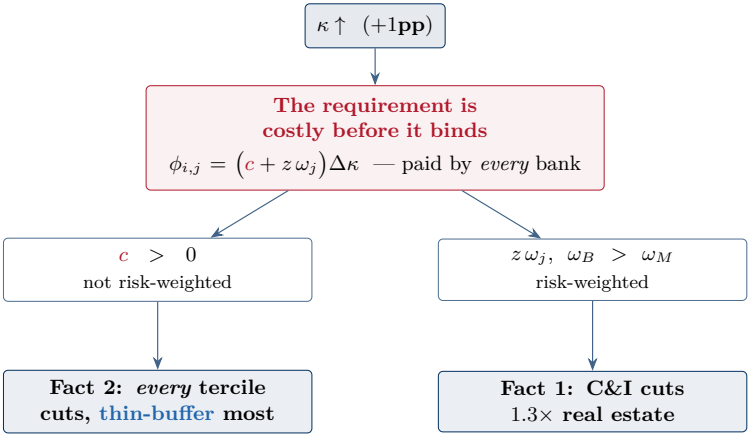
So the requirement is costly *before* it binds:

$$\phi_{i,j}(\kappa) \;=\; \big(c + z\,\omega_j\big)\,\max(\kappa - \bar{\kappa}_i,\, 0)$$

- The gradient says this cost *exists*; the *sectoral* split says what it is made of: c is the same for every kind of loan, $z\,\omega_j$ depends on the kind

Facts recap What c and z are

The Mechanism



One term, both facts: c sets the level, $z\,\omega_j$ the sectoral split.

Model

What the Model Adds

The local projections give responses. The model gives three things they cannot:

- **Incidence** — the burden *per dollar lent*, by buffer position
- **A price** — the welfare cost of a tightening, as a share of credit-market surplus
- **Counterfactuals** — a uniform requirement, or flat risk weights: regimes the data never ran

Model: Environment

Dynamic, partial equilibrium, heterogeneous-agent.

- Continuum of infinitely lived, risk-neutral banks; equity state $e_{i,t}$
- Two sectors—real estate ($j = M$) and C&I ($j = B$)—with inverse loan demand $r_{j,t} = \theta_j - \eta_j L_t^j$

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- Retail deposits are limited — the marginal dollar comes from wholesale funding $d_{i,t}^W$, at quadratic cost $\frac{\gamma_i}{2}(d_{i,t}^W)^2$
- $\gamma_{\text{small}} \gg \gamma_{\text{large}} \to 0$: small banks face steep wholesale costs — this is where buffer heterogeneity comes from

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- $\gamma_{\text{small}} \gg \gamma_{\text{large}} \to 0$: small banks face steep wholesale costs — this is where buffer heterogeneity comes from
- One aggregate disturbance: κ_t follows an AR(1); bank i faces $\kappa_{i,t} = \kappa_t +$ its legal tier add-on
- Two occasionally binding constraints: regulatory capital, and dividend non-negativity

Market clearing Stylized model

The Dynamic Bank Problem: The Problem

Banks choose lending and dividends to maximise the discounted dividend stream:

$$V(e_{i,t}, A^M_{i,t-1}, A^B_{i,t-1}; \Omega_t) = \max_{\{A^M_{i,t},\, A^B_{i,t},\, c_{i,t}\}} \Big\{ c_{i,t} + \beta\, \mathbb{E}_t\, V\big(e_{i,t+1}, A^M_{i,t}, A^B_{i,t}; \Omega_{t+1}\big) \Big\}$$

Ω_t is the aggregate state: the current requirement and the two sectoral credit aggregates that clear the loan markets.

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Subject to the balance-sheet identity — wholesale funding is *residual*, not chosen:

$$d^W_{i,t} = A^M_{i,t} + A^B_{i,t} - e_{i,t} - d_{i,C}$$

and to the law of motion for equity, which carries returns, costs and the levy into $e_{i,t+1}$.

Equity law of motion

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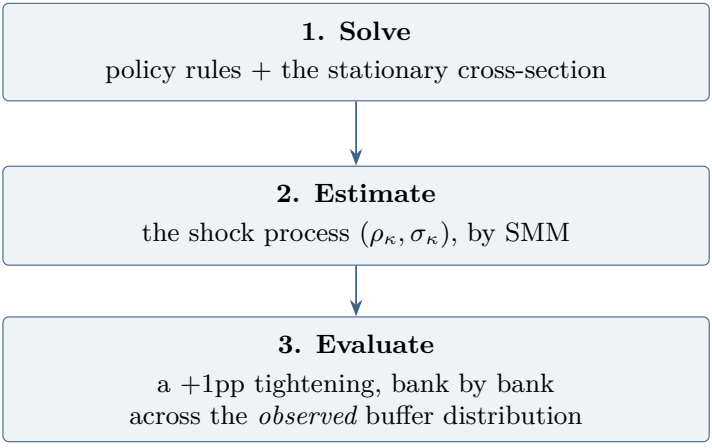
Equity law of motion

and to two occasionally binding constraints:

$$\underbrace{\omega_M A^M_{i,t} + \omega_B A^B_{i,t} \le \frac{e_{i,t}}{\kappa_{i,t}}}_{\text{regulatory capital}} \iff \underbrace{D_{i,t} \ge 0}_{\text{the buffer of Fact 2}}, \qquad \underbrace{c_{i,t} \ge 0}_{\text{no equity issuance}}$$

Euler equation · Solution method · Calibration

How the Model Is Used



SMM Estimation Strategy

Estimate: $\theta = (\rho_\kappa, \sigma_\kappa)$ — persistence and volatility of the AR(1) capital-requirement process

Targets: 16 within-bank LP coefficients
C&I and RE responses at $\mathcal{H} = \{1, 2, 4, 6, 8, 12, 16, 20\}$

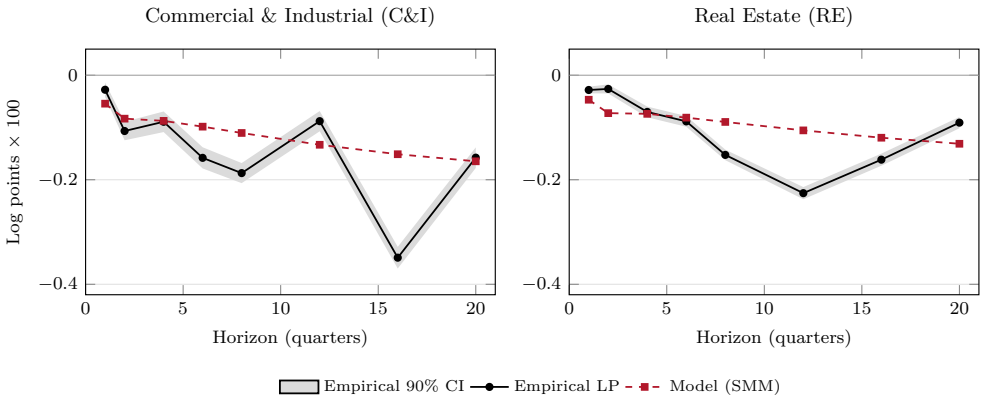
Identification, in words:

- $\rho_\kappa \leftarrow$ how slowly the IRF decays at medium horizons
- $\sigma_\kappa \leftarrow$ how large the impact response is
- Two parameters, sixteen moments

LP specification | Weighting and algorithm | Calibration

Estimation: Model vs. Data

The 16 targeted moments: sectoral credit responses, model against data.



$\hat{\rho}_\kappa = 0.9987$: a tightening is not reversed — 97% of it is still in place at horizon 20.

Moment fit | Magnitudes | Robustness

A Prediction, Not a Fit

Targeted — the 16 moments in the previous figure:

sectoral credit responses, C&I and real estate, at eight horizons

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Not targeted — which banks cut most:

Thin-to-thick response ratio	data (90% CI)	model
C&I	1.32 [1.01, 1.62]	**1.23**
Real estate	1.41 [1.14, 1.67]	**1.18**

▸ **The ratio:** banks re-sorted each quarter into *lagged*-buffer terciles, the same LP run in each; thin tercile's mean b_h over $h = 1$–20, divided by the thick tercile's. Above 1: thin-buffer banks cut more.

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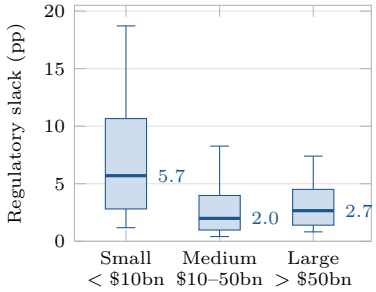
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- The model was fitted to *how much* credit falls — never to *which banks* cut most.

Which solve Precision

Who Bears It? Incidence Across Banks

The regulation sorts by size. The mechanism sorts by buffer.



Buffers vary as much *within* each statutory tier as across them.

In the model, a +1pp tightening:

- **Thin-buffer banks:** 32% of lending, 34% of the burden
- **Thick-buffer banks:** 35% of lending, 33% of the burden

Per dollar of total lending, thin-buffer banks bear 13% more than thick-buffer banks.

How it's measured | Tier mix | By tier | Over time | Buffer by size

Conclusion

Who bears the burden of bank capital regulation?

- **Which sectors:** risk weights concentrate the contraction in C&I — 1.31× the real-estate response
- **Which banks:** the buffer, not size — in the data, thin-buffer banks cut 1.32× the thick-buffer response, within C&I
- **And the law does not see it:** the statute sorts by size; the burden sorts by distance to the requirement

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Design ⇒ **who pays**. *Level* ⇒ the aggregate cost.

1.4–2.4% of credit-market surplus per +1pp, gross of the stability benefits of capital.

Thank you

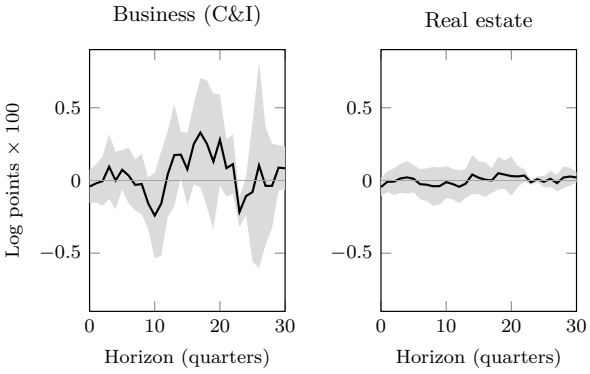
Who Bears the Burden of Bank Capital Regulation?

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Fact 1: Sectoral Asymmetry in Credit Responses

Identification: Narratively identified capital-requirement shocks (Drechsel and Miura, 2025); within-bank local projections



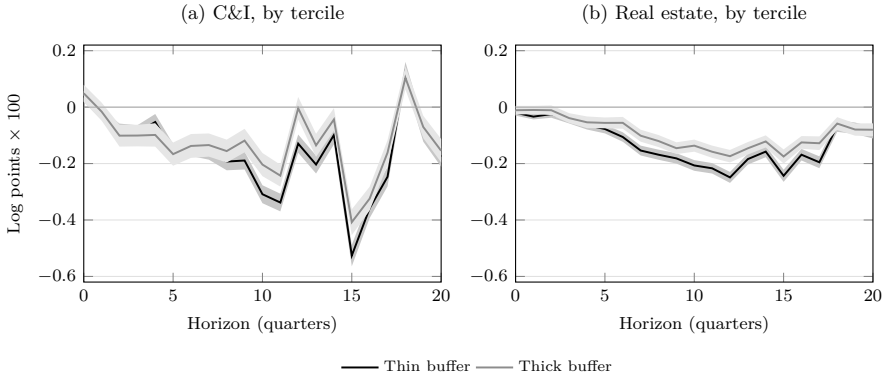
Scale Deleveraging Multiplier

$$\frac{\partial \text{RWA}}{\partial A^B} = \omega_B > \omega_M = \frac{\partial \text{RWA}}{\partial A^M}$$

Each dollar of C&I carries twice the regulatory cost per dollar lent — paid *whether or not* the constraint binds.

All four | Identification | Interpretation

Fact 2: Thin-Buffer Banks Cut More



Banks sorted by lagged buffer $D_{i,t-1}$.

Thin-buffer banks contract 1.32× more in C&I — and 1.41× in real estate.

Details Precision

Interpreting the Sectoral Asymmetry

Why C&I contracts more than RE:

- **Asset Substitution Margin:** when the constraint binds, banks shed high-ω assets (C&I, $\omega_B = 1$) to free the most required capital per dollar cut

- **Scale Deleveraging Multiplier:** $\partial \text{RWA}/\partial A^B = \omega_B > \omega_M = \partial \text{RWA}/\partial A^M$
Each dollar of C&I cut yields twice the regulatory relief of a mortgage cut

- **And away from the frontier:** $\partial \phi_{i,j}/\partial \kappa = c + z\,\omega_j > 0$, so the same ordering holds for *every* bank, constrained or not — which is why the asymmetry is not confined to the <3% at the frontier

- **Consumer credit unresponsive:** high interest margins provide sufficient cash-flow buffers to absorb the regulatory cost without volume rationing

Result: the sectoral composition of the contraction is determined by the risk-weight schedule, not by loan demand.

Back

SMM Moment Fit

Sector	Horizon h	$\hat{b}_h$ (data)	s.e.	b_h (model)	t
C&I	1	-0.028	0.008	-0.054	+3.20
	2	-0.107	0.011	-0.083	-2.20
	4	-0.089	0.012	-0.087	-0.14
	6	-0.158	0.012	-0.098	-4.85
	8	-0.187	0.012	-0.110	-6.57
	12	-0.088	0.012	-0.133	+3.84
	16	-0.349	0.013	-0.151	-15.73
	20	-0.158	0.012	-0.165	+0.58
RE	1	-0.028	0.004	-0.047	+4.87
	2	-0.026	0.005	-0.073	+8.80
	4	-0.070	0.006	-0.074	+0.59
	6	-0.088	0.006	-0.081	-1.15
	8	-0.152	0.006	-0.089	-10.29
	12	-0.226	0.007	-0.106	-17.02
	16	-0.162	0.007	-0.120	-5.97
	20	-0.091	0.007	-0.131	+6.16
Hansen $J = 1822.15$, dof $= 14$, $\chi^2_{0.05} = 23.68$, $p = 0.000$					

Signs correct for all 16 moments; sectoral ordering reproduced at every horizon. $J = 1822$ (14 d.f.) reflects the high precision of the bank panel—the economically relevant misfit is in the trough depths.

Back

Identification: Capital-Requirement Shocks

Shock series: Drechsel and Miura (2025)

- Narrative + sign-restriction approach
- Captures unanticipated changes in the capital-requirement stance
- Available 2002Q1–2023Q2; estimation sample capped at 2019Q4

Local Projection specification:

$$x_{t+h} - x_{t-1} = \alpha_h + \beta_h \, \text{shock}_t^{\text{Cap}} + \sum_{p=1}^{6} \Psi_{h,p} \, x_{t-p} + \varepsilon_{t+h}$$

- **Aggregate:** OLS, Newey–West (lag $h + 6$)
- **Bank-level:** bank FE, clustered by cert
- Consumer credit placebo: no response $\Rightarrow$ supply, not demand

Back

The Buffer Gradient: Details

- Thin-buffer banks cut C&I 1.32× more than thick-buffer banks (90% CI [1.01, 1.62]); 1.41× in real estate, 1.37× pooled [1.13, 1.61] — from a joint cluster-robust covariance across the separate tercile regressions

- The risk-weight mechanism implies the *same* gradient in both sectors, and the restriction is not rejected: difference +0.090 (s.e. 0.186), $t = 0.48$

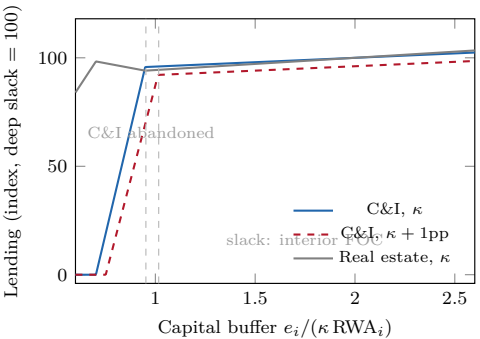
- Freezing tercile membership at each bank's pre-sample average slack **reverses** the gradient, to 0.67 in C&I — it is the buffer a bank holds when the shock arrives that matters, not a fixed bank type

Horizon profile, cohort power, and the continuous-interaction specification: see backup.

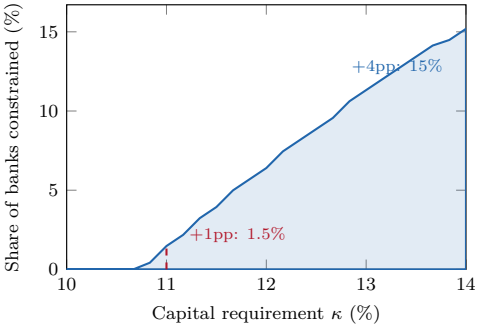
Back

Why a Global Non-Linear Method

(a) Policy functions are kinked at the frontier



(b) Transmission depends on the mass at the kink



Both panels are the *exact* static solution — the problem is quadratic with a linear constraint, so it solves in closed form in each KKT regime.

What the kink implies Back

Why a Global Non-Linear Method

- The capital constraint binds occasionally, so the policy function is **kinked**: left of the frontier lending is pinned by equity, right of it the interior FOC holds and lending is flat in equity
- A +1pp tightening moves the kink *right* — banks that were unconstrained become constrained
- So the aggregate response is not a fixed elasticity: it depends on the **mass sitting near the kink**. +1pp constrains 1.5% of banks, +4pp constrains 15%

The figure Back

Backup: Neural-Network Solution

Challenge: occasionally binding constraints ⇒ kinked policy functions.

Approach: parameterise policy with a multi-layer perceptron (MLP):

$$\begin{bmatrix} \hat{A}^M_{i,t} \\ \hat{A}^B_{i,t} \\ \hat{c}_{i,t} \\ \hat{\lambda}_{i,t} \\ \hat{\mu}_{i,t} \end{bmatrix} = \mathcal{A}(S_{i,t}; \Theta) \qquad S_{i,t} \equiv (e_{i,t}, A^M_{i,t-1}, A^B_{i,t-1}, \Omega_t)$$

$\Omega_t \equiv (\kappa_t,\, L^M_t,\, L^B_t)$: the current requirement and the two sectoral credit aggregates.

Network Method Non-convexity Accuracy What it pins down Back

Backup: Solution Method — Implementation

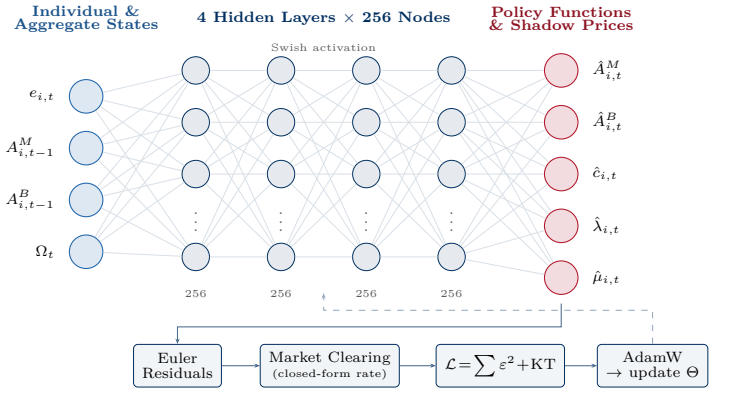
How the policy is trained.

- **Structural loss:** squared Euler residuals + soft KT penalty
- **Market clearing:** imposed analytically via closed-form equilibrium rate
- **Architecture:** 4 hidden layers $\times$ 256 nodes, Swish activation
- **Optimizer:** AdamW with cosine annealing

The Karush–Kuhn–Tucker conditions enter as a soft penalty rather than being imposed by construction; the multiplier uses the cubed García–Zangwill substitution, which makes complementarity hold identically and keeps the map twice differentiable at the kink.

Back to network Back to solution

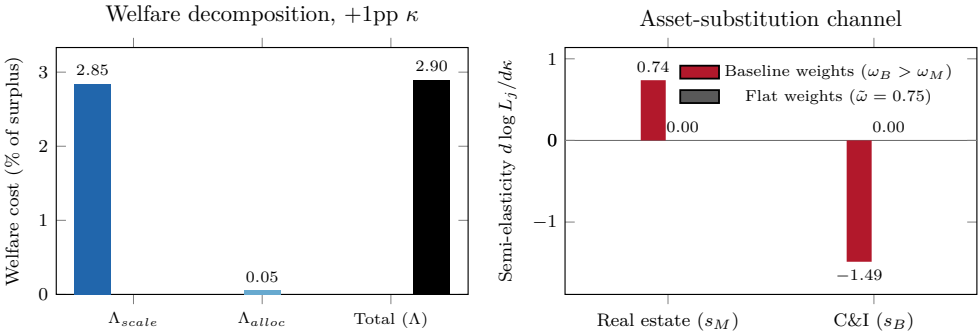
Neural-Network Architecture



Training: ∼6h on NVIDIA A100 · Cosine annealing schedule · Euler errors $< 10^{-3}$ across state space

Implementation Back

Gross Cost and Decomposition



Left: CEV decomposition. Right: constrained-regime semi-elasticities.

Counterfactuals: neither a uniform $\tilde{\kappa}$ nor flattened risk weights changes the aggregate burden — but both change **who bears it**.

Counterfactuals | Details | CEV framework | Magnitudes in context

Gross Cost: Numbers

Experiment	Quantity	Value
CEV of +1pp κ tightening (eqs. (??)–(??))		
Total welfare cost	Λ	1.421%
scale channel	Λ_{scale}	1.415%
allocation channel	Λ_{alloc}	0.004%
additivity check	$\Lambda_{scale} + \Lambda_{alloc}$	1.419%
Credit contraction (total / RE / C&I)	ΔL	-3.09% / -3.02% / -3.50%
CEV from the franchise value V (discounted dividends)		
Total welfare cost	Λ_V	2.404%
scale channel	$\Lambda_{V,scale}$	2.392%
allocation channel	$\Lambda_{V,alloc}$	0.026%
additivity check	sum	2.419%
CF1: uniform vs. size-differentiated κ (eq. (??))		
Differentiated rule	Λ	1.236%
Uniform $\tilde{\kappa} = 0.1246$	Λ	1.239%
CF2: flat risk weights $\tilde{\omega} = 0.75$ (eq. (??))		
Baseline sectoral asymmetry	$s_B - s_M$	-2.224
Flat-weight sectoral asymmetry	$s_B - s_M$	-0.000

Credit-market surplus: $\Lambda = 1.4\%$. Franchise value: $\Lambda_V = 2.4\%$. Scale channel accounts for >99% of the cost. Magnitudes are *directional*: robust in sign and decomposition, sensitive to the solved policy. The cross-sectional gradient is not.

Back

Policy Counterfactuals

CF1: Uniform vs. size-differentiated κ

- Replace the step function with a uniform $\tilde{\kappa} = 0.1246$ raising the same aggregate capital
- Gross cost: 2.680% vs. 2.682% $\Rightarrow$ size-tailoring redistributes but barely reduces the aggregate burden

CF2: Risk-weight flattening

- Set $\tilde{\omega}_B = \tilde{\omega}_M = 0.75$ (average), hold κ fixed
- Sectoral asymmetry $s_B - s_M$ collapses from -2.22 to 0.00
- Aggregate credit contraction essentially unchanged

Level of requirements $\Rightarrow$ aggregate cost.
Design (risk weights, size tiers) $\Rightarrow$ incidence.

Details Back

Backup: Counterfactual Definitions

CF1 — uniform vs. size-differentiated requirements. Replace the statutory step function $\kappa_i(\text{Scale}_i)$ with the single rate $\tilde{\kappa}$ that raises the *same aggregate capital* in the stationary distribution μ^*:

$$\tilde{\kappa} \;\text{ s.t. }\; \int \tilde{\kappa}\, e_i \,\mathrm{d}\mu^* \;=\; \int \kappa_i(\text{Scale}_i)\, e_i \,\mathrm{d}\mu^*.$$

So the experiment holds the *level* of regulation fixed and varies only its *design*: any difference in Λ is pure incidence.

CF2 — flat risk weights. Set $\omega_M = \omega_B = \tilde{\omega} = 0.75$, removing the risk-weight wedge. The test is whether the two sectors then deleverage proportionally:

$$\frac{\partial \tilde{L}^B_{t+h}}{\partial \varepsilon^{\kappa}_t} \;\approx\; \frac{\partial \tilde{L}^M_{t+h}}{\partial \varepsilon^{\kappa}_t} \quad \forall\, h.$$

	Λ	sectoral asymmetry $s_B - s_M$
Differentiated (baseline)	1.236%	-2.224
Uniform $\tilde{\kappa} = 0.1246$	1.239%	—
Flat weights $\tilde{\omega} = 0.75$	—	-0.000

Design redistributes the burden; it barely moves the aggregate. Flattening the weights eliminates the sectoral asymmetry entirely.

Back

Backup: Related Literature

Quasi-experimental credit supply

- Aiyar, Calomiris, and Wieladek (2016), Gropp, Guettler, and Saadi (2020), De Jonghe, Dewachter, and Ongena (2020), Jimenez et al. (2017), Conti, Nobili, and Signoretti (2023)
- Here: the same identified shock is traced across sectors *and* across the buffer distribution, then mapped into a structural cost.

Deposit franchise and regulatory incidence

- Drechsler, Savov, and Schnabl (2017)
- The deposit franchise shapes not only monetary transmission but the incidence of *capital* regulation: what disciplines the buffer is the *marginal* funding source.

Quantitative macro-banking

- Corbae and Erasmo (2021), Begenau (2020), Elenev, Landvoigt, and Van Nieuwerburgh (2021), Mendicino et al. (2020)

Back

Backup: Reading c and z

$$\phi_{i,j}(\kappa) \;=\; \big(c + z\,\omega_j\big)\,\max(\kappa - \bar{\kappa}_i,\,0)$$

- $z = 0.021$ is **risk-weighted**: a Modigliani–Miller premium, $\approx 10\%$ annualised equity–debt spread
- $c = 0.148$ is **not** risk-weighted and seven times larger — what a **leverage requirement** imposes. U.S. banks face one; the model's constraint is risk-based only
- Both are set on the two sectoral aggregates; *both* buffer gradients are untargeted

The leverage-requirement reading is a conjecture about the *source* of the non-risk-weighted term. The data identify its *magnitude* either way.

Back

Backup: SMM Weighting and Algorithm

Weighting matrix

$\mathbf{W} = \hat{\Sigma}^{-1}$, block-diagonal, estimated by cluster bootstrap of the bank-level local projections (100 replications)

Algorithm — for each candidate $\theta = (\rho_\kappa, \sigma_\kappa)$

1. Re-solve the model globally with the neural network
2. Simulate a panel impulse path under the candidate process
3. Compute model-implied LP coefficients on the simulated panel
4. Evaluate $(\hat{m} - m(\theta))' \mathbf{W} (\hat{m} - m(\theta))$

Each evaluation requires a full re-solve, which is what makes the global solution method the binding cost of estimation.

Hansen J rejects. With ≈4,800 banks the moments are basis-point precise, so the test weights economically trivial misses heavily: sign, sectoral ordering and hump shape all match; trough depth does not.

Validation is on the *ratio*, not the level: σ_κ is a free scale, so the depth of any model response is fitted by construction. This is why the untargeted test is stated as a thin-to-thick ratio.

Back

Backup: The Within-Bank LP Specification

One panel regression per sector and horizon, on bank-quarter data:

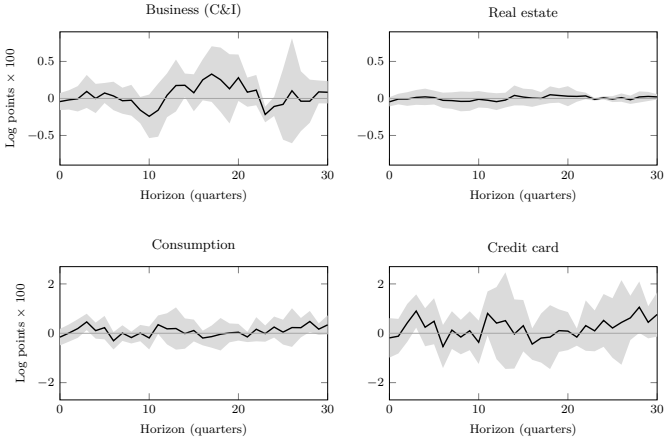
$$\underbrace{\log L^{j}_{i,t+h} - \log L^{j}_{i,t-1}}_{\text{cumulative response}} = \alpha_i + b^{j}_{h}\,\varepsilon^{\kappa}_{t} + \sum_{p=1}^{6} \gamma_p' X_{i,t-p} + u_{i,t+h}$$

- α_i **is what "within-bank" means:** each bank is compared with *its own* average, so permanent differences across banks drop out
- ε^{κ}_{t} is the narrative capital-requirement shock — one aggregate series, identical for every bank, non-missing in 72 quarters
- X = log RE lending, log C&I lending, log equity, GDP growth, fed funds; six lags each. Series winsorised 1/99; standard errors clustered by bank

What comes out is one number, not one per bank. The fixed-effects slope is exactly a weighted average of the bank-by-bank slopes, $b^{j}_{h} = \sum_i w_i \hat{\beta}_i$, with w_i each bank's share of within-bank shock variance — here *observation counts*, never dollars. So b^{j}_{h} is the response of the **average bank**, not of aggregate credit.

Sample: banks under \$10bn through 2019Q4. Estimated at $h = 0, \ldots, 30$; the SMM targets keep $\mathcal{H} = \{1, 2, 4, 6, 8, 12, 16, 20\}$ for two sectors = 16 moments. Bands are ± 1.645 standard errors.

Backup: All Four Loan Categories



C&I contracts most, real estate less, and consumption and credit-card lending do not respond at any horizon. The two unresponsive categories are the demand placebo: they are fully risk-weighted but funded against high interest margins, so a capital tightening need not be absorbed through volume.

Back

Backup: Summary of Empirical Facts

Fact	Finding
1. Sectoral asymmetry	C&I contracts ∼a third more than RE within banks (risk-weight wedge)
2. Buffer gradient	Thin-buffer banks cut 1.32× more; with terciles frozen at pre-sample slack the gradient *reverses* (0.67)
3. Placebo	Consumption and credit-card lending unresponsive at all horizons
4. Robustness	Results stable to 2023Q2 extension, pandemic controls

Back

The Dynamic Bank Problem (II): The Euler Equation

Optimal lending in sector j satisfies:

$$\beta\,\mathbb{E}_t\left[(1+\mu_{i,t+1})\left(r_{j,t}-\varepsilon^j_{i,t+1}-\psi_j A^j_{i,t}\underbrace{-r_{W,t}-\gamma_i d^W_{i,t}}_{\text{marginal funding}}-\phi_{i,j}(\kappa_{i,t})\right)\right]=\lambda_{i,t}\,\omega_j$$

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- $\lambda_{i,t}$: shadow cost of the capital constraint — zero whenever $D_{i,t}>0$
- $\mu_{i,t}$: multiplier on $c_{i,t}\geq 0$ — internal funds are worth more than one when the bank would cut dividends below zero and cannot

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- $\mu_{i,t}$: multiplier on $c_{i,t}\geq 0$ — internal funds are worth more than one when the bank would cut dividends below zero and cannot
- $\phi_{i,j}$: per-unit cost of regulatory capital, paid *whether or not* the constraint binds (bank-specific via $\bar{\kappa}_i$)
- $\omega_B>\omega_M \Rightarrow$ higher regulatory cost per unit C&I $\Rightarrow$ C&I contracts more

Where these terms come from Back

Backup: Where the Euler Terms Come From

Law of motion for equity. Every term in the Euler equation is a derivative of this line.

$$e_{i,t+1} = (1-\delta_e)\, e_{i,t} + \sum_{j\in\{M,B\}} \left[(r_{j,t} - \varepsilon^j_{i,t+1}) A^j_{i,t} - \tfrac{\psi_j}{2}(A^j_{i,t})^2 - \phi_{i,j}(\kappa_{i,t})\, A^j_{i,t} \right]$$
$$- r_{d,t}\, d_{i,C} - r_{W,t}\, d^W_{i,t} - \tfrac{\gamma_i}{2}(d^W_{i,t})^2 - c_{i,t} - \tau_t e_{i,t}$$

Wholesale funding is residual, not chosen: $d^W_{i,t} = A^M_{i,t} + A^B_{i,t} - e_{i,t} - d_{i,C}$

$\varepsilon^j_{i,t+1}$: idiosyncratic credit loss, realised after lending, so the net return per dollar is $r_{j,t} - \varepsilon^j_{i,t+1}$

ψ_j: quadratic operating cost — marginal cost $\psi_j A^j_{i,t}$ rises with portfolio size, which pins the long-run sectoral mix

$r_{d,t}$, $r_{W,t}$: retail deposit and wholesale funding rates

δ_e: quarterly equity depreciation (operating costs, franchise decay)

τ_t: systemic insurance levy on equity, funding entrant capital injections

Back to Problem (I) Back to Euler (II)

Backup: Funding Frictions and the Target Buffer

Proposition 1. In the stationary equilibrium, a bank with funding friction γ_i and core-deposit capacity $\bar{d}_{i,C}$ targets wholesale funding

$$d_i^{W\star} = \frac{\rho + \delta_e + \tau - r_W}{\gamma_i},$$

and holds equity $\bar{e}_i$ satisfying $\bar{e}_i = A_i(\bar{e}_i) - \bar{d}_{i,C} - d_i^{W\star}$. Consequently:

1. the target capital ratio $\bar{e}_i/\mathrm{RWA}_i$ is strictly increasing in γ_i;
2. there is a threshold γ^*: banks with $\gamma_i \leq \gamma^*$ target the regulatory frontier *exactly* — zero voluntary buffer — while banks with $\gamma_i > \gamma^*$ target a strictly positive buffer;
3. the elasticity of target equity to the requirement is ≈ 1 for $\gamma_i \leq \gamma^*$ and ≈ 0 for $\gamma_i > \gamma^*$.

▸ (i) is the *costlier funding, thicker target buffer* claim; (ii) is why a few banks sit exactly at the frontier and most do not; (iii) says a requirement change moves target equity only for the banks already at the frontier

Back Funding capacity

Backup: Slack on Funding Capacity

Dep. var: Slack (pp)	(1) Pooled	(2) Between	(3) Within
Funding Capacity	-19.44^{***}	-25.97^{***}	-2.27^{**}
	(1.73)	(1.95)	(1.05)
Log Assets	-2.80^{***}	-2.70^{***}	-6.27^{***}
	(0.09)	(0.10)	(0.16)
Quarter FE / Bank FE	Yes / No	—	Yes / Yes
N	935,234	16,117	935,056

The between estimate (-25.97) is the cross-sectional regularity; the within estimate (-2.27) is small because the funding mix barely moves at quarterly frequency. Size enters separately and negatively throughout, so this is not size in disguise.

Back

Stylized Model: Environment

Static, partial equilibrium. Continuum of risk-neutral banks $i \in [0, 1]$.

- Two asset classes: mortgages (A_i^M, $\omega_M = 0.50$) and C&I (A_i^B, $\omega_B = 1.00$)
- Liabilities: core deposits $d_{i,C}$ (cheap, limited) + wholesale $d_{i,W}$ (quadratic friction $\frac{\gamma_i}{2} d_{i,W}^2$)
- $\gamma_{\text{small}} \gg \gamma_{\text{large}} \to 0$: small banks face steep wholesale costs
- Regulatory constraint: $e_i \geq \kappa(\omega_M A_i^M + \omega_B A_i^B)$

Capital buffer:

$$D_i = e_i - \kappa(\omega_M A_i^M + \omega_B A_i^B) \geq 0$$

Banks with high $\gamma_i \Rightarrow$ large endogenous D_i (self-insurance).

FOCs Back

First-Order Conditions and the Sectoral Channel

FOC for asset class $j \in \{M, B\}$:

$$r_j - r_W - \psi_j A_i^j - \gamma_i (A_i^M + A_i^B - e_i - d_{i,C}) = \lambda_i \kappa \omega_j$$

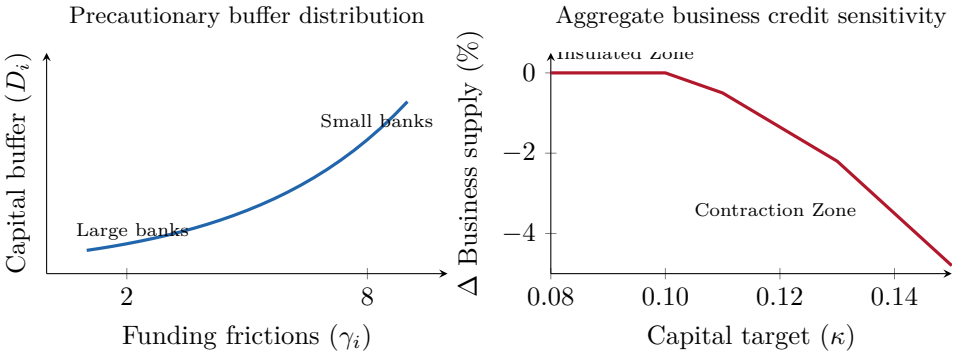
When $\lambda_i > 0$ (constraint binds):

- **Asset Substitution Margin:** $\left|\partial A_i^B / \partial \kappa\right| > \left|\partial A_i^M / \partial \kappa\right|$ because $\omega_B > \omega_M$
- **Scale Deleveraging Multiplier:** $\partial \text{RWA} / \partial A^B = \omega_B = 1 > 0.5 = \omega_M \Rightarrow$ cutting C&I frees 2× the regulatory capital per dollar

Cross-section: for low-γ banks, λ_i stays low $\Rightarrow$ small response. For high-γ banks whose buffer is breached, λ_i spikes $\Rightarrow$ acute, non-linear contraction in C&I.

Insulated zone Back

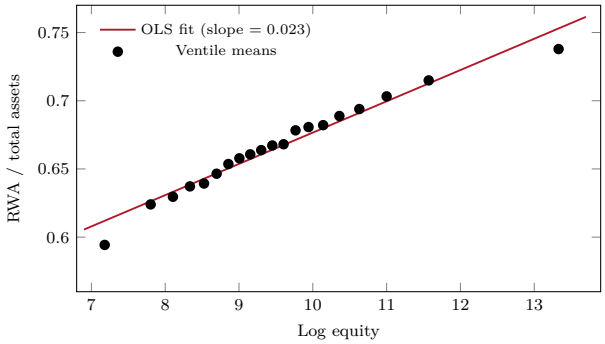
The Insulated Zone



When shocks are small-to-moderate, well-buffered banks absorb them internally. Only shocks large enough to breach the buffer trigger non-linear credit contractions.

Back

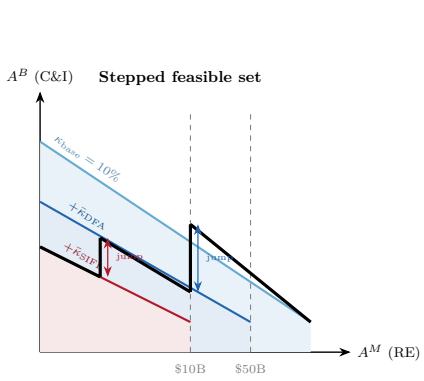
The Regulatory Wedge



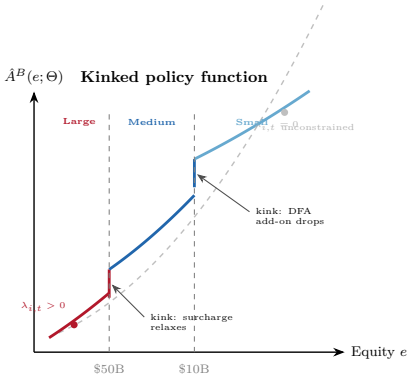
RWA/total-assets ratio rises with bank equity ⇒ larger banks hold higher average risk weights; portfolio composition drives the regulatory position of small banks.

Back

Why Neural Networks? The Non-Convexity Challenge



$\omega_M A^M + \omega_B A^B \leq e/\kappa_i(\text{Scale})$
Non-convex: frontier jumps at size thresholds



Policy $\hat{A}^B(e;\Theta)$ has kinks at each tier boundary $\Rightarrow$ perturbation methods fail

Size-tiered surcharges create a stepped feasible set (left) and kinks in optimal policy (right). Standard perturbation around the steady state cannot capture these discontinuities—motivating a global neural-network solution.

Back

Calibration

Parameter	Description	Value	Source
β	Discount factor	0.99	Standard
δ_e	Equity depreciation	0.001	Stationarity
ω_M, ω_B	Risk weights	0.50, 1.00	Basel III
κ_{base}	PCA minimum	10%	Regulatory
$\bar{\kappa}_{\text{DFA}}$	DFA add-on	1.5%	Enhanced prudential
$\bar{\kappa}_{\text{SIFI}}$	Large-bank add-on	2.5%	Stylised
γ_{small}	Wholesale funding cost	0.077	Funding structure
γ_{large}	Wholesale funding cost	$\to 0$	Funding structure
ψ_M	RE adjustment cost	0.007	RE share target
ψ_B	C&I adjustment cost	0.05	Preset
c	Cost of capital, *not* risk-weighted	0.148	Sectoral aggregates
z	Cost of capital, per unit RWA	0.021	Jointly with c
η_M, η_B	Demand slopes	external	Lit. semi-elasticities
$\sigma_{\varepsilon,M}$	Idiosync. shock (RE)	0.001	Charge-off data
$\sigma_{\varepsilon,B}$	Idiosync. shock (C&I)	0.004	Charge-off data

Demand sensitivity
Back to the problem
Back to SMM

Gross Cost: Framework

Consumption-equivalent variation (CEV):

$$V^{\kappa}(\mu^*;\, \kappa^* + 0.01) = V^*((1-\Lambda)\,\mu^*;\, \kappa^*)$$

$\Lambda > 0$ = fraction of steady-state surplus households would forgo to avoid the tightening.

Decomposition:

$$\Lambda = \underbrace{\Lambda^{\text{alloc}}}_{\text{sectoral distortion}} + \underbrace{\Lambda^{\text{scale}}}_{\text{credit contraction}}$$

- Two metrics: credit-market surplus and franchise value (discounted dividends)
- **Caveat:** gross cost—no stability benefits, partial equilibrium

Surplus figure

Back

Backup: Robustness to Sample Extension

Re-estimating small-cohort LPs on the full shock series (2002Q1–2023Q2):

Treatment	Mean C&I response	Mean RE response
Baseline (2002–2019)	−0.145%	−0.105%
Pandemic dummy × shock	−0.142%	−0.106%
Exclude pandemic shocks	−0.142%	−0.106%

Target moments are essentially unchanged.

Sensitivity to ψ_B Back

Backup: Sensitivity to ψ_B

Constrained-regime sectoral asymmetry $s_B - s_M$ across the admissible range of $\psi_B \in (0.014, 0.142)$:

ψ_B	$s_B - s_M$
0.02	-3.5
0.05 (baseline)	-2.2
0.10	-0.8
Flat weights (any ψ_B)	0.0

The asymmetry is a risk-weight phenomenon; ψ_B governs only its amplitude.

Back

Backup: Precision of the Buffer Gradient

Terciles are estimated as separate local projections, so the *difference* needs a joint covariance: a bank contributes scores to several tercile blocks over its life, and the twenty horizons use overlapping windows.

	Thin	Thick	Ratio (90% CI)
C&I	−0.1718	−0.1307	1.32 [1.01, 1.62]
Real estate	−0.1354	−0.0964	1.41 [1.14, 1.67]
Pooled	—	—	1.37 [1.13, 1.61]

The cross-sector restriction is not rejected. The risk-weight mechanism implies the same gradient in both sectors: measured difference +0.090 (s.e. 0.186), $t = 0.48$, across 9,801 banks.

To the facts

To the prediction

Backup: What the Solved Policy Does and Does Not Pin Down

The model is solved by training a policy network.

Stable across solved policies. The welfare cost. With the cost of regulatory capital active, Λ lies between 1.16% and 1.42% across solved lending from 0.21× to 1.05× the calibration target — it tracks the *proportional* contraction ($\approx -3\%$), not the level.

Not stable. The impulse-response magnitudes, and the buffer gradient when it is recomputed *from* a solved policy rather than from the static problem.

So the gradient we report is the exact one. The static problem is quadratic with a linear constraint, so it solves in closed form in each KKT regime: C&I 1.32 against 1.32 in the data. The trained policy returns 1.02–1.06 on the same calibration — it smooths the kink at the frontier, where its residuals are largest — and neither added capacity nor a four-fold longer schedule closes the gap.

What collapses Λ to 0.1% is not a low solved level but *removing* $\phi_{i,j}$: without it total credit moves -0.25% rather than -3.1%. That is a different model, not a numerical sensitivity — and it is the sharpest evidence that $\phi_{i,j}$, not the capital constraint, generates the welfare cost.

To the prediction To solution method

Backup: Market Clearing

Inverse demand in each sector:

$$r_{j,t} = \theta_j - \eta_j L_t^j, \qquad j \in \{M, B\}$$

Clearing against the solved policies:

$$\tfrac{1}{N} \textstyle\sum_{i=1}^{N} \hat{A}_{i,t}^{j}(\mathbf{S}_{i,t}; \Theta) \;=\; L_t^j(r_{j,t})$$

Substituting gives a closed-form rate — imposed analytically at every forward pass, so there is *no numerical inner loop*:

$$r_{j,t}^{*} = \theta_j - \eta_j \cdot \tfrac{1}{N} \textstyle\sum_{i=1}^{N} \hat{A}_{i,t}^{j}(\mathbf{S}_{i,t}; \Theta)$$

Demand calibration Back

Backup: Market Clearing

The two schedules are pinned by sample averages and a semi-elasticity:

$\eta_j = 1/(S_j \bar{L}^j)$ with S_j the demand semi-elasticity; $\theta_j = \bar{r}_j + \eta_j \bar{L}^j$, so the schedule passes through the sample average.

	mortgages (M)	C&I (B)
η_j	6.497e-8	3.773e-7
θ_j	0.0462	0.0847
$\bar{r}_j$	0.0154	0.0169
$\bar{L}^j$ (USDmm)	473,578	179,618
implied S_j	32.5	14.8

Sensitivity Clearing Back

Backup: Sensitivity to the Demand Calibration

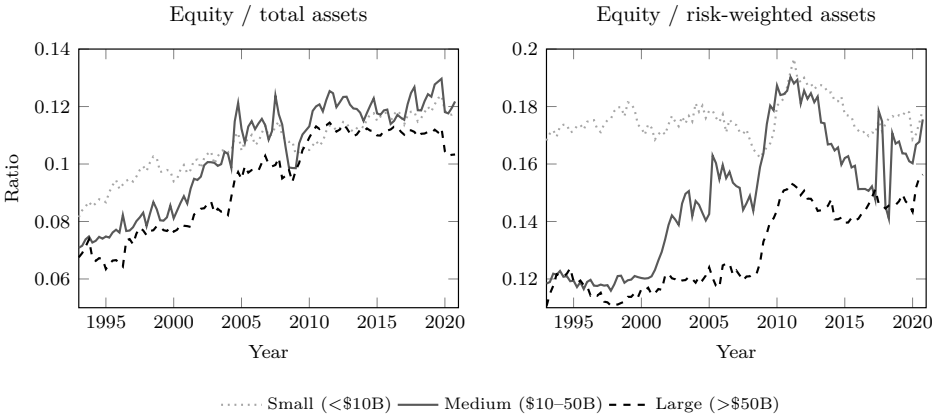
η_j enters neither the bank's problem nor the reduced-form estimates, so the lending responses are *exactly* invariant to it. The welfare *level* is not, because the surplus integral is taken under the demand curve.

S_M	S_B	Λ
32.5 (implemented)	14.8	1.42%
16.3	7.4	0.95%
2.0 (cited targets)	4.0	0.27%

The externally cited estimates measure objects adjacent to a loan-demand schedule — a house-price elasticity, and a loan-*granting* probability against the policy rate — rather than the schedule itself.

Back

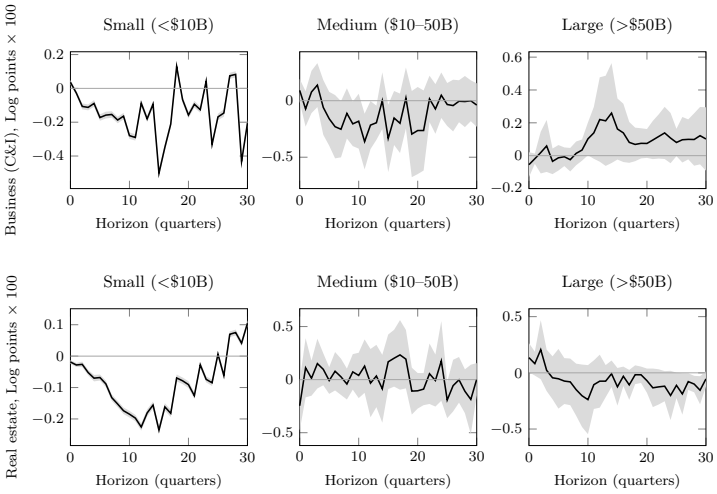
Backup: Leverage vs. Regulatory Capital Ratios



Small banks appear more capitalised on an unweighted basis, but the advantage compresses under risk-weighting.

Back

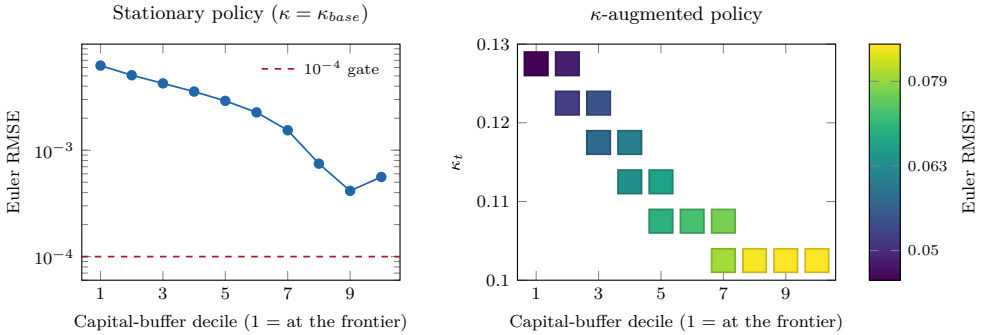
Backup: Size Cohort Impulse Responses



Small-bank panel precisely estimated; large-bank cohort (≈40 institutions) is not. The buffer distribution, not size per se, drives the cross-section.

Back

Backup: Solution Accuracy

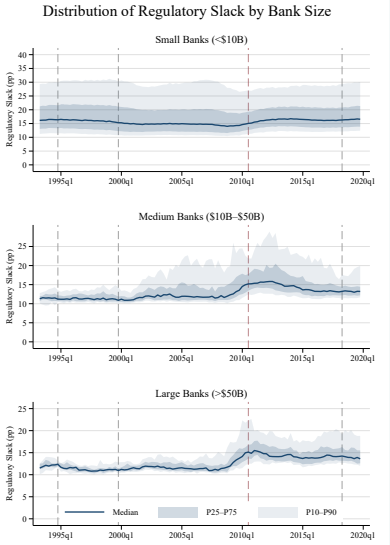


All deciles below 10^{-2}. Frontier spike reflects the kink in policy functions at the occasionally binding constraint.

Train vs. test | Ergodic coverage

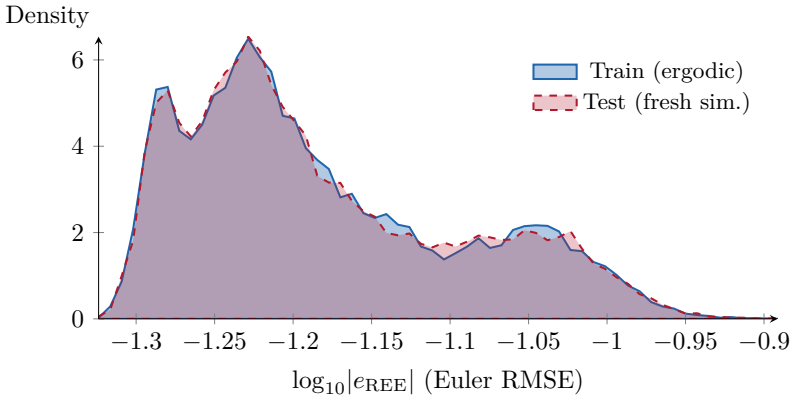
Back

Backup: Slack Distribution by Size, 1993Q2–2019Q4



Shaded bands: P10–P90 (light) and P25–P75 (dark). Solid line: median.
Dashed vertical lines mark Riegle-Neal (1994Q4), GLBA (1999Q4), Dodd-Frank (2010Q3, highlighted), and EGRRCPA (2018Q2).
Sample: quarterly bank panel, 1993Q2–2019Q4. Source: US Call Reports.

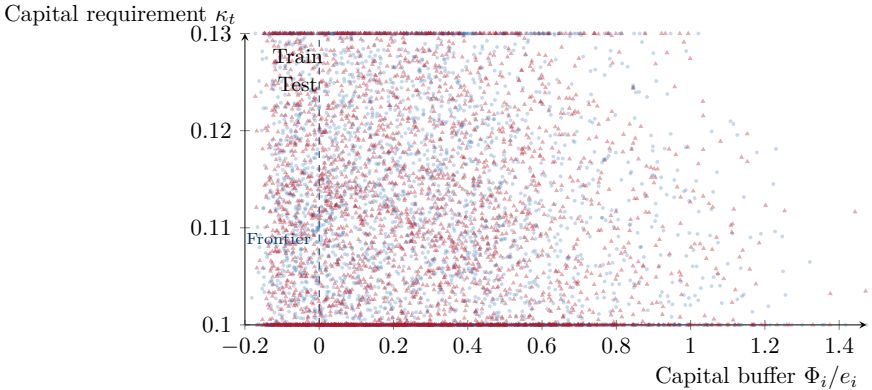
Backup: Euler Errors, Training vs. Test



Distribution of $\log_{10}|e_{\mathrm{REE}}|$ on the training panel (seed 0) and an independent test panel (seed 42), 20,000 banks each after 200 periods of burn-in. Median residual -1.21 in *both* samples and no fattening of the test tail: the network generalises to unseen states rather than memorising the training trajectory.

Back

Backup: Ergodic-Set Coverage



Joint distribution of the capital buffer Φ_i/e_i and the requirement κ_t, training against test simulation. The test cloud lies entirely inside the training cloud's convex hull, so the policy is evaluated only where it was solved — extrapolation beyond the training support is the most common failure mode of neural-network approximations.

Back

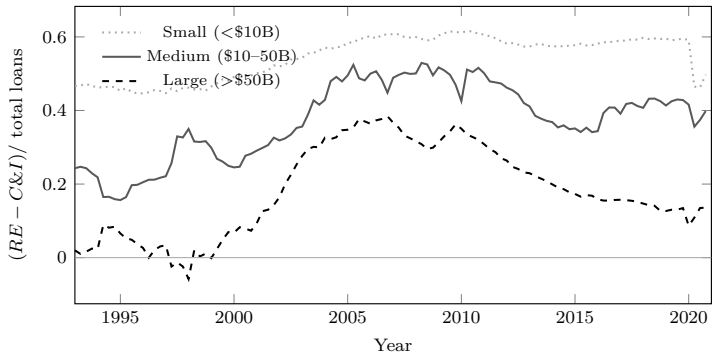
Backup: Magnitudes in Context

Study	Design / setting	Effect of +1pp required capital
Aiyar, Calomiris, and Wieladek (2016)	UK bank-level requirement changes, 1998–2007	Lending growth 6.5–7.2pp lower
Mésonnier and Monk (2015)	EBA 2011–12 capital exercise, euro area	Annualised loan growth 1.2–1.6pp lower
De Jonghe, Dewachter, and Ongena (2020)	Pillar-2 decisions, Belgian credit register, 2013–15	Corporate credit growth $\approx$ 0.2–0.4pp per quarter lower
Conti, Nobili, and Signoretti (2023)	Narrative BVAR, Italy (EBA/CA episodes)	Loan supply -1.5% after 2 years per +40bp Tier 1 ($\approx -3.8\%$ per pp, linearised)
This paper (model)	SMM-estimated policy, permanent +1pp κ	C&I -3.5%, RE -3.0%, total credit -3.1% (stationary comparison)
This paper (LPs)	Within-bank LPs scaled through the SMM mapping (one shock unit $\mapsto \hat{\sigma}_\kappa = 4.0$bp of κ)	Trough responses $\approx -8.8\%$ (C&I, h=16) and $\approx -5.7\%$ (RE, h=12); out-of-sample linear extrapolation

Model's -3.5% C&I / -3.0% RE / -3.1% total per +1pp sits squarely inside the quasi-experimental range. These magnitudes are conditional on the solved policy; the cross-sectional gradient is not.

To estimation To gross cost

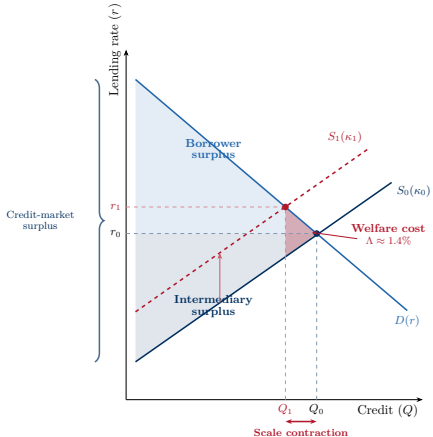
Backup: Asset Gap



Difference between RE and C&I lending as a share of total loans, by size cohort, 1993–2020.

Back

Welfare: Credit-Market Surplus

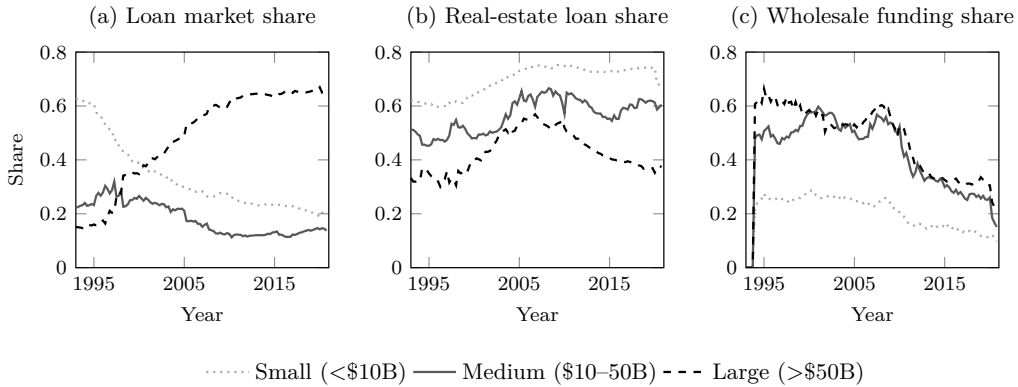


Virtually all of the welfare cost flows through the contraction in the **scale** of intermediation ($\Lambda^{\text{scale}} \gg \Lambda^{\text{alloc}}$)

Borrower + intermediary surplus in the loan market. The red triangle is the welfare cost of a +1pp κ tightening; virtually all flows through **scale** contraction ($\Lambda^{\text{scale}} \gg \Lambda^{\text{alloc}}$).

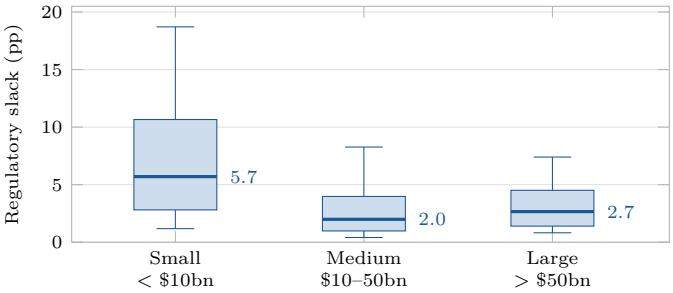
Back

Backup: Market Concentration and Asset Specialisation



Asset gap Back

Backup: Buffer Heterogeneity Cuts Across Size Tiers



Buffers vary as much *within* each statutory tier as across them.

Full distribution | Size-cohort IRFs | To funding capacity | To who bears it

Backup: How Incidence Is Measured

Burden = dollars of lending lost. **Incidence** = a group's share of those dollars, against its share of lending:

$$\text{concentration}_g \;=\; \frac{\sum_{i\in g}\Delta A_i \big/ \sum_i \Delta A_i}{\sum_{i\in g} A_i \big/ \sum_i A_i}, \qquad A_i \equiv A_i^M + A_i^B$$

Buffer tercile	lending	burden	**conc.**	contraction	binding
Thin	32.1%	34.1%	**1.06**	−0.127%	3.4%
Middle	32.9%	33.0%	1.00	−0.120%	0.0%
Thick	35.1%	33.0%	**0.94**	−0.113%	0.0%

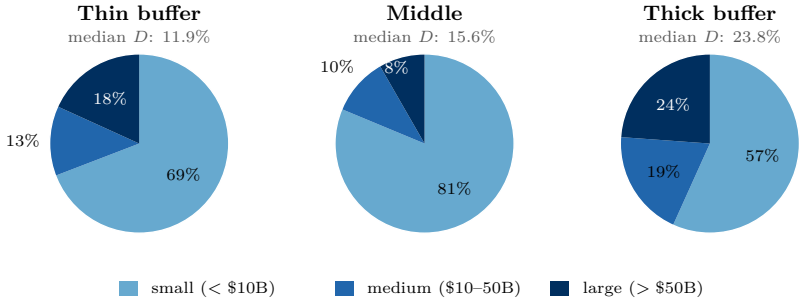
Thin against thick: $1.06/0.94 = 1.13$ — the "13% more per dollar lent" of the previous slide. Thin-buffer banks bear only 6% more than their *own* lending share.

▸ **Scale-free:** every term is a share, so σ_κ, ρ_κ and the shock size cancel — incidence is a property of the policy function across the cross-section, not of the shock process

4,000 bank-quarters, +4bp tightening, static problem (closed form in each KKT regime), representative scale.

Back

Backup: Who Sits in Each Buffer Tercile?

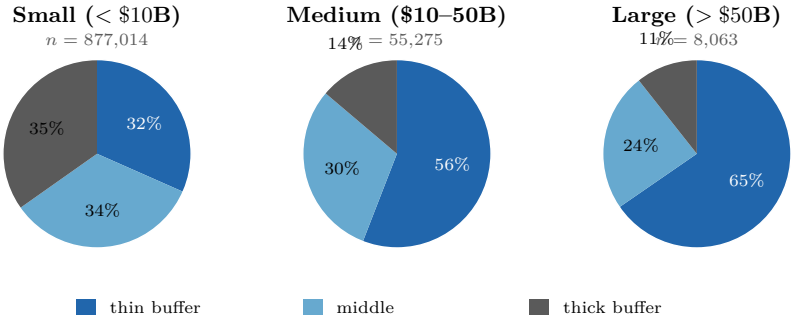


Share of each tercile's *assets* held by banks in each statutory tier. Call Reports through 2019Q4, 20,000 bank-quarters.

- The size does not sort buffers monotonically: the statutory tier is not a proxy for distance to the requirement

Back

Backup: Buffer Position Within Each Size Tier

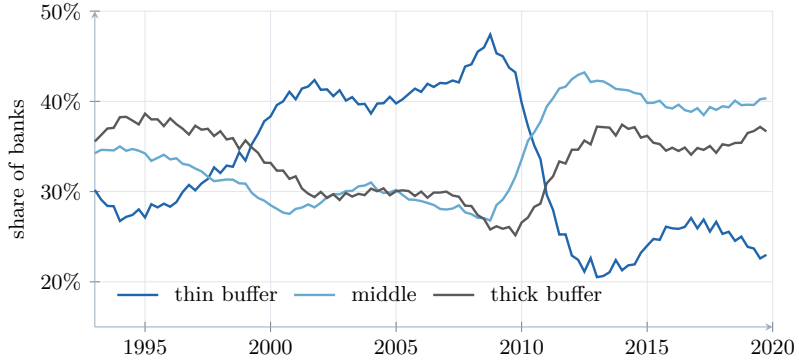


Share of each *size tier's* bank-quarters falling in each buffer tercile. Cutoffs fixed at the pooled 13.65% / 18.47% capital ratio.

- Read down, not across: small banks split almost evenly (32/34/35), but 65% **of large bank-quarters are thin-buffered** and only 11% thick

Back

Backup: Buffer Composition Over Time



Tercile cutoffs held *fixed* at the pooled 13.65% / 18.47%, so the shares move; 108 quarters, 1993Q1–2019Q4.

- Tercile membership is a *state*, not a bank type: the same cutoffs classify a very different population before and after 2010

Back

The Regulatory Framework

Size cohorts by total assets (Dodd-Frank/EGRRCPA[1] thresholds):

- Community, $<$ \$10B — baseline κ
- Regional, \$10–50B — $\kappa + 1.5$pp (Dodd-Frank)
- Large, $>$ \$50B — $\kappa + 2.5$pp (enhanced standards)

Concentration Regulatory wedge Regulatory tiers Back

[1] Economic Growth, Regulatory Relief, and Consumer Protection Act (2018), which raised the automatic enhanced-prudential-standards threshold from \$50B to \$250B. The \$50B tier governs almost the whole 1994–2019 sample.

Backup: Regulatory Tiers, Dodd-Frank and 2019

Over the estimation sample (Dodd-Frank, 2010–2018)

Tier	Trigger	Main requirements
Community	$< \$10$B	Baseline risk-based minima; PCA well-capitalised
Regional	\$10–50B	Company-run stress tests; risk committee
Large	$> \$50$B	Enhanced prudential standards: CCAR, supervisory stress tests, liquidity standards, resolution plans

After EGRRCPA (2018) and the tailoring rule (October 2019)

Category	Trigger	Main requirements
I	The eight U.S. G-SIBs	G-SIB surcharge, eSLR, TLAC, AOCI in capital, full LCR/NSFR
II	$\geq \$700$B assets *or* $\geq \$75$B cross-jurisdictional	Full LCR/NSFR; AOCI in capital
III	$\geq \$250$B assets *or* $\geq \$75$B weighted STWF, nonbank assets, or off-balance-sheet exposure	Reduced LCR/NSFR (85%) unless STWF $\geq \$75$B; AOCI opt-out
IV	\$100–250B, none of the above	Biennial supervisory stress test; no eSLR; AOCI opt-out

- Note: category depends on **five** indicators, not size alone — one of them is **weighted short-term wholesale funding**, the margin γ_i represents.

G-SIB surcharge Back

Backup: The G-SIB Surcharge (Category I)

The surcharge is the *higher* of two scores, assigned in 50bp buckets:

- **Method 1** (Basel): size, interconnectedness, *substitutability*, complexity, cross-jurisdictional activity — 20% each
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Two consequences for this paper:

- The two margins of the model—wholesale funding and risk weights—enter *one* regulatory indicator, multiplied together
- Substituting C&I ($\omega_B = 1$) into real estate ($\omega_M = 0.5$) lowers RWA, so it *raises* the funding indicator and the surcharge — the opposite sign to the mechanism

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March 2026 proposal: index the method-2 coefficients to growth and inflation (they are nominal, so surcharges have drifted up); drop the RWA denominator; reweight STWF to 20% from ≈30%; 10bp buckets. Estimated effect: −3.8% G-SIB CET1.

Back to tiers Back to Data

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